```
from scipy import stats
import math

mu_0=200
sample_std=15
sample_mean=190
n=40
alpha=0.05

t_stat=(sample_mean-mu_0)/(sample_std/math.sqrt(n))

df=n-1

t_critical=stats.t.ppf(alpha,df)

p_value=stats.t.cdf(t_stat,df)

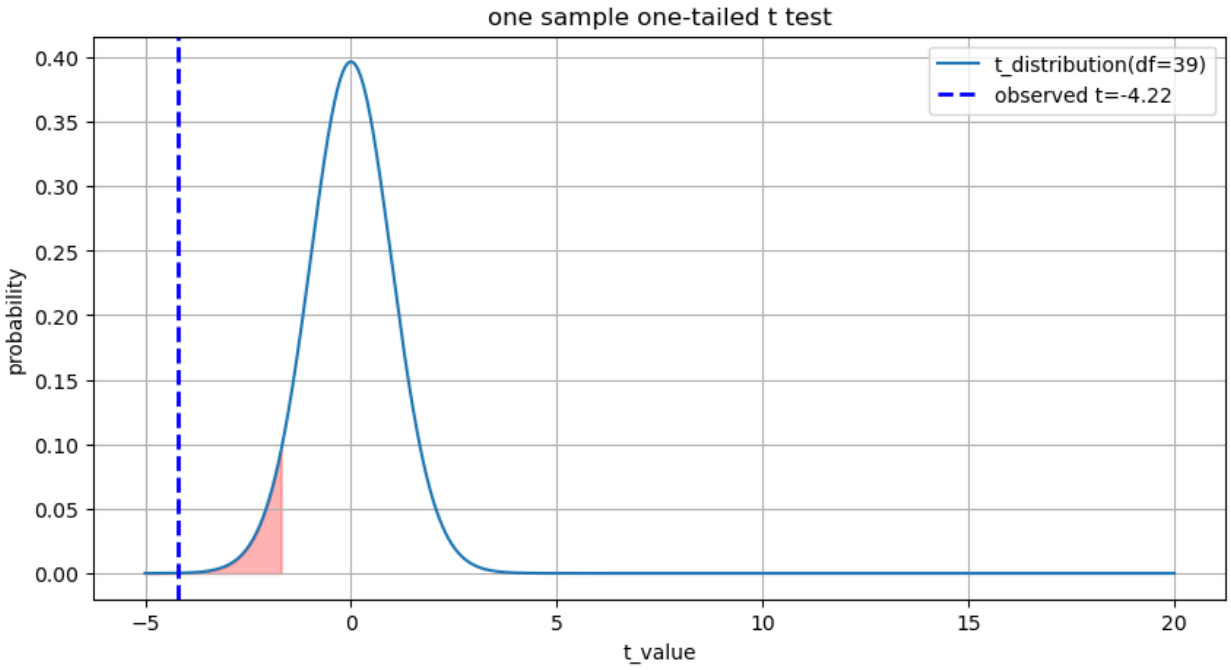
print(f"t-statistics:{t_stat:.4f}")
print(f"critical t-value(one_tailed,alpha=0.05):{t_critical:.4f}")
print(f"p_value:{p_value:.4f}")
```

```
t-statistics:-4.2164
critical t-value(one_tailed,alpha=0.05):-1.6849
p_value:0.0001
```

```
if p_value<alpha:
    print("reject the null hypothesis: the new recipe has
significantly fewer calories")
else:
    print("fail to reject the null hypothesis :no significant
reduction in calories")
```

```
reject the null hypothesis: the new recipe has significantly fewer
calories
```

```
import numpy as np
import matplotlib.pyplot as plt
x=np.linspace(-5,20,1000)
y=stats.t.pdf(x,df)
plt.figure(figsize=(10,5))
plt.plot(x,y,label=f"t_distribution(df={df})")
plt.fill_between(x,0,y,where=(x<=t_critical),color='red',alpha=0.3)
plt.axvline(t_stat,color='blue',linestyle='--',linewidth=2,label=f"obs
erved t={t_stat:.2f}")
plt.title("one sample one-tailed t test")
plt.xlabel("t_value")
plt.ylabel("probability")
plt.legend()
plt.grid(True)
plt.show()
```


one sample one-tailed t test
t_distribution(df=39)
observed t=-4.22
probability
t_value
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−5
0
5
10
15
20